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Deliverable 3 — Use-Case Flow Specification

Academic Elective Bidding & Allocation System

Use Case: UC-02 — Submit Elective Bids

Primary Actor: Student

Supporting Actor: University Authentication System

1. Preconditions

1. The student has logged into the system successfully.
2. The elective bidding period is open.
3. The student is eligible to take part in the elective bidding process.
4. The student has 100 bidding credits available.

2. Postconditions

If the submission is successful:

- The student's elective preferences and bidding credits are saved.
- The submitted bids are ready to be considered during the allocation process.
- The student sees a confirmation message.

If the submission is not successful:

- The invalid bids are not submitted.
- The student's bidding credits are not changed until a valid submission is made.

3. Main Success Scenario

1. The student opens the elective bidding section.
2. The system shows the electives available to the student.
3. The student chooses the electives they want to bid for.
4. The student arranges the selected electives in order of preference.
5. The student distributes their 100 bidding credits among the selected electives.
6. The system adds up the credits assigned to the selected electives.
7. The system checks that the total number of credits is not more than 100.
8. The system checks whether the student has completed the prerequisites for the selected electives.
9. The system shows the student a summary of the bids for review.
10. The student confirms the submission.
11. The system saves the student's preferences and credit distribution.
12. The system shows a message confirming that the bids were submitted successfully.
13. The use case ends.

4. Alternate Flow — Student Uses More Than 100 Credits

7a1. At Step 7, the system finds that the student has assigned more than 100 credits.

7a2. The system does not allow the bid submission to continue.

7a3. The system tells the student that the total bidding credits cannot be more than 100.

7a4. The student changes the credit distribution.

7a5. The system calculates the total again.

7a6. Once the total is 100 or less and the required prerequisites are satisfied, the student can continue.

7a7. The flow continues from Step 9.

5. Additional Validation

- If the student has not completed a prerequisite for a selected elective, the system should flag that elective, prevent it from being submitted, and inform the student that the prerequisite has not been satisfied.